Corporate Venture Investment Intelligence & Target Selection

Final Project Documentation — Version 2.0

Status: Project complete — SEC/XBRL ETL, MySQL/SQL screening and a three-page Power BI dashboard delivered; broader investment factors are future scope

1. Document Purpose

This document is the final project playbook for the Corporate Venture Investment Intelligence & Target Selection portfolio project. It records the business objective, data sources, ETL methodology, financial analysis, investment-intelligence logic, SQL analysis, Power BI dashboard, validation, errors encountered and their resolutions, final conclusions and limitations.

The Jupyter Notebook contains the implementation details. This document captures the professional project story, decisions, evidence, and interview-ready explanations.

2. Business Problem

Corporate venture and strategic investment teams may need to screen many public companies before allocating detailed due-diligence resources. This project builds a data-driven screening workflow that consolidates public financial information, evaluates financial performance and direction, identifies potential warning signals, and prioritizes companies for further human review.

The solution is a decision-support screening framework, not an autonomous investment adviser.

3. Project Objectives

- Build a reliable, source-backed financial dataset.
- Automate public-company financial extraction using SEC EDGAR/XBRL data.
- Create reusable Python ETL rather than relying on manual extraction.
- Standardize company-year financial information.
- Calculate growth, profitability, cash-generation and directional indicators.
- Create transparent investment-intelligence flags.
- Persist the analytical output in MySQL.
- Use SQL to compare, rank and classify potential targets.
- Present the results through a professional Power BI dashboard.

4. Company Universe

Company	Ticker	Financial Period
Etsy, Inc.	ETSY	FY2023–FY2025
Shake Shack Inc.	SHAK	FY2023–FY2025
Crocs, Inc.	CROX	FY2023–FY2025
Boot Barn Holdings, Inc.	BOOT	FY2023–FY2025
Duolingo, Inc.	DUOL	FY2023–FY2025

5. Project Roadmap & Status

Phase	Scope	Status
Phase 1	SEC/XBRL data acquisition	Completed
Phase 2	Reusable financial ETL	Completed
Phase 3	Financial analytical dataset	Completed
Phase 4	Investment intelligence	Completed
Phase 5	SQL screening & candidate classification	Completed
Phase 6	Broader investment insights	Future scope
Phase 7	Power BI dashboard	Completed
Phase 8	Portfolio packaging & interview story	Completed

6. Data Sources & Technology Stack

- SEC EDGAR / data.sec.gov Company Facts XBRL API — primary financial source.
- Python — requests and pandas for extraction, transformation, QA and analysis.
- Jupyter Notebook / Anaconda — development environment.
- MySQL / MySQL Workbench — persistent analytical storage and SQL analysis.
- PyMySQL + SQLAlchemy — Python-to-MySQL connectivity.
- Power BI Desktop — visualization and decision-support layer, connected to MySQL; DAX measures, calculated columns and calculated tables.
- HTML prototypes — three layout prototypes used to agree the dashboard design before building it in Power BI.
- Power BI theme (CVI Dark Executive, JSON) — one consistent dark executive look across all pages.

7. Financial ETL Methodology

- Configure company identifiers, fiscal years and required XBRL concepts.
- Retrieve SEC Company Facts JSON.
- Identify appropriate US-GAAP concepts and USD units.
- Filter annual observations for FY2023–FY2025.
- Resolve repeated comparative observations and retain the appropriate filing record.
- Standardize monetary values to USD thousands.
- Build a consistent company-year financial structure.
- Calculate growth, margins, changes and directional signals.
- Run QA checks for year coverage, duplicates, nulls and structure.
- Build investment-intelligence flags.
- Load the final company-level screening output into MySQL.

8. Key Financial Metrics

The financial foundation uses revenue, gross profit where available, operating income, net income and operating cash flow. Derived measures include revenue growth, gross margin, operating margin, net margin, CFO margin, changes in margins, CFO-to-net-income relationships, and operating-income change.

9. Cross-Company Financial Results

Company	Revenue Growth 23–25	Operating Margin	Net Margin	CFO Margin	Operating Income Change

					($M)
DUOL	95.36%	13.07%	39.91%	37.38%	+148.83
SHAK	32.90%	4.32%	3.16%	15.38%	+56.59
BOOT	15.29%	12.52%	9.47%	7.72%	+7.57
ETSY	4.92%	9.23%	5.65%	24.05%	-13.63
CROX	1.98%	3.70%	-2.01%	17.58%	-887.27

10. Directional Investment-Intelligence Framework

The screening framework evaluates five directional signals: growth, operating margin, net margin, cash generation and operating income. Each company receives positive and negative direction counts. The net-direction score is calculated as positive signals minus negative signals.

Company	Positive	Negative	Net Score	Classification
DUOL	5	0	+5	Strong Candidate
SHAK	5	0	+5	Strong Candidate
BOOT	3	2	+1	Watchlist
ETSY	1	4	-3	Weak Candidate
CROX	0	5	-5	Weak Candidate

11. Earnings vs Cash-Flow Diagnostic

Company	Flag	Net Margin	CFO Margin
DUOL	Aligned positive	39.91%	37.38%
SHAK	Divergent periods present	3.16%	15.38%
BOOT	Divergent periods present	9.47%	7.72%
ETSY	Divergent periods present	5.65%	24.05%
CROX	Aligned positive	-2.01%	17.58%

The earnings/cash-flow flag is a diagnostic rather than an automatic investment judgment. It is intended to identify periods where earnings and operating cash-flow behavior deserve additional investigation.

12. SQL Analysis Completed

- Compared companies by positive and negative signal counts.
- Calculated net-direction score.
- Compared revenue growth and profitability.
- Compared operating, net and cash-flow margins.
- Analyzed operating-income dollar change.
- Reviewed improving and declining financial indicators.
- Reviewed earnings versus operating-cash-flow consistency.
- Created candidate classifications.
- Produced a final SQL-ready company-level screening output.

13. Final SQL Screening Result

The final SQL output combines candidate classification with the most important supporting financial evidence. The resulting ranking is:

Rank Group	Company	Classification	Net Score	Key Interpretation
1	DUOL	Strong Candidate	+5	Strong growth,

				profitability and cash-generation direction
1	SHAK	Strong Candidate	+5	Positive direction across all five screening signals
3	BOOT	Watchlist	+1	Positive growth but mixed profitability direction
4	ETSY	Weak Candidate	-3	Growth remains positive but several profitability/cash signals weakened
5	CROX	Weak Candidate	-5	Broad negative directional pattern under current framework

14. Financial Screening Interpretation

DUOL emerged as the strongest financial screening candidate. It achieved 95.36% revenue growth from FY2023 to FY2025, a 13.07% operating margin, 39.91% net margin, 37.38% CFO margin and a $148.83M increase in operating income.

SHAK also achieved a +5 net-direction score with 5 positive and 0 negative signals. Its FY2023–FY2025 revenue growth was 32.90%, with positive directional movement across operating margin, net margin, CFO margin and operating income.

BOOT is a Watchlist candidate because growth and cash-generation indicators are positive, while operating and net margins have declined.

ETSY and CROX are classified as Weak Candidates under the current financial-direction framework. This does not mean they are universally unattractive investments; it means their current financial signals are weaker under the defined screening rules.

15. MySQL Implementation

Database: investment_intelligence. Main screening table: investment_screening.

The final screening DataFrame contains 5 rows and 20 analytical columns. The dataset was successfully loaded into MySQL using a SQLAlchemy engine with the PyMySQL driver.

The direct pandas-to-SQL attempt using a raw PyMySQL connection produced a compatibility error. The workflow was corrected by using SQLAlchemy, which is the appropriate connection layer for pandas to_sql.

A second table, financial_data (company-year grain, 13 columns, key = ticker + fiscal_year), holds 15 rows (5 companies × FY2023–FY2025). It exists because Power BI trend charts need fiscal_year and yearly values, which the company-level screening table does not contain. Shake Shack gross profit and Crocs operating cash flow are stored as NULL because they are not available, not as zero.

The MySQL Workbench Table Data Import Wizard loaded only 6 of the 15 rows. The load was replaced with an explicit SQL INSERT script (TRUNCATE followed by 15 rows with NULLs), which loaded all 15 rows and is the repeatable load method for this table.

16. Validation & QA

- Final screening DataFrame shape: 5 × 20.
- Five unique companies present.
- Expected candidate classifications produced.
- Positive signal count across companies: 14.
- Negative signal count across companies: 11.
- MySQL investment_screening table contains 5 company records.
- SQL outputs were cross-checked against the Python screening results.
- Power BI reconciliation: 5 companies; Strong 2, Watchlist 1, Weak 2; total positive/negative signals 14 / 11; ranking table and net-score chart match the SQL results.
- Rank measure uses tie-aware logic (companies with a higher Net Score + 1), giving ranks 1, 1, 3, 4, 5.
- Dashboard cross-checks: the company slicer filters the KPI cards and the detail charts through the ticker relationship; missing values (Shake Shack gross profit, Crocs operating cash flow) display as blank, not zero.

17. Error & Resolution Log

Issue	Cause	Resolution	Interview Lesson
Etsy duplicate _x/_y columns	Repeated merges created duplicate metric names.	Inspected columns and retained the validated metric fields.	Always inspect merge outputs before downstream calculations.
revenue_df missing after kernel restart	Jupyter kernel reset cleared in-memory variables.	Rebuilt the missing upstream object and reran dependent cells.	Notebook state must not be treated as permanent storage.
Boot Barn SEC 404	Initial CIK was incorrect.	Corrected the SEC identifier and reran extraction.	Validate regulatory identifiers before automation.
investment_intelligence_flags KeyError	The expected column was not present during partial notebook recovery.	Inspected the actual columns and reran the appropriate upstream logic.	Debug dependencies from actual object state, not assumptions.
pandas to_sql / PyMySQL error	Raw DBAPI connection was not supported cleanly by the installed pandas workflow.	Moved to a SQLAlchemy engine.	Use the correct abstraction layer for pandas database operations.
SQLAlchemy host parsing error	Connection string was malformed because of a special character in the credential.	Corrected/encoded the connection string.	Credentials containing URL-reserved characters must be encoded.
Large notebook state loss	Kernel restart removed all variables from	Ran the complete notebook from a clean state and fixed errors	A production-style pipeline

	memory.	sequentially.	should be reproducible from a clean environment.
Power BI: Cannot find table investment_screening	Power BI imported the table as "investment_intelligence investment_screening" (schema + table name).	Used the full table name in DAX, or renamed the table in the Data pane.	Check the exact object names Power BI creates on import before writing DAX.
Power BI: classification column not found	The MySQL table holds direction counts and flags but no classification column; the classification rule lived in the SQL query output.	Created Net Score and Classification calculated columns in DAX (Strong ≥ +3, Watchlist 0 to +2, Weak < 0 — thresholds inferred from current results and to be aligned with the SQL rule).	Materialize business rules such as classification in the database so every tool uses the same rule.
DAX syntax error: "Total" is incorrect	Two measures were pasted into one formula bar.	Created one measure per New measure step.	Each DAX measure is created and saved separately.
Strong Candidates card showed 3 instead of 2	The measure condition did not isolate Strong Candidates; the KPI cards summed to 6 for 5 companies.	Corrected the measure so the cards reconcile to 5 / 2 / 1 / 2.	Reconcile KPI totals against known results before trusting a visual.
Rank column showed 1, 1, 1, 4, 5	Tie handling gave BOOT the same rank as the tied leaders.	Replaced the measure with "number of companies with a higher Net Score + 1" (expected 1, 1, 3, 4, 5); re-verify after applying.	Test ranking logic with ties.
Field parameter: syntax error on closing brace	A full formula was pasted over the existing one, leaving a duplicate closing brace.	Removed the extra brace.	Edit only the part that changes in generated DAX.
Metric field parameter gave a blank chart	Likely caused by manually editing the field-parameter definition; a normal column in the same chart displayed correctly, so the chart itself was fine.	Replaced it with a Metric Options table (DATATABLE) plus a SWITCH measure, Selected Metric Value, with Order as the sort column.	Isolate a fault by swapping one field at a time; prefer a simple, controllable design.
Company-year table missing in MySQL	Only investment_screening existed; trend charts need ticker, fiscal_year and yearly values.	Created financial_data (13 columns) from the Financial Data sheet.	Confirm the data grain supports every planned visual before building.
Import Wizard loaded 6 of 15 rows	The wizard stopped part-way (likely on rows with blank numeric cells; exact cause not confirmed).	Used a SQL INSERT script with explicit NULLs; verify COUNT(*) = 15.	Prefer scripted loads for small datasets: explicit and repeatable.
Metric buttons and company	Text slicers sort	Sort by column (Order) for the	Control sort

bars in changing order	alphabetically, and the chart re-sorts by value whenever the metric changes.	buttons; a Company Order calculated column (RANKX on Net Score, then revenue growth) is used to fix the Strong → Weak order.	order explicitly with a numeric helper column.
Company Detail cards ignored the company slicer	The yearly table (financial_data) and the screening table had no relationship, and the slicer used the yearly table's full company names.	Related financial_data[ticker] to the screening table's company (many-to-one, single direction) and built the slicer on the screening table's company.	Check keys and relationships in Model view before building visuals; the key values must match exactly.
Gross margin line missing for Shake Shack	Not a bug: Shake Shack gross profit is NULL in the dataset and the measure deliberately returns blank.	Kept it blank (not zero) and explained it on the page.	Show missing data as missing, and tell the reader why.
Trend chart titled "%GT" with a single line	A "Show value as: percent of grand total" setting was applied to the measure, and the Legend field was empty so all companies were summed.	Set Show value as to No calculation and put ticker in the Legend.	If values look like shares of 100%, look for an unintended quick calculation.
Margins chart drawn as a 100% stacked column	Wrong visual type for margins over time.	Changed it to a line chart with fiscal_year on the axis and the four margin measures.	Match the chart type to the question: lines for change over time.
Trend chart showed only 2024 and 2025	Revenue growth has no FY2022 baseline, so FY2023 is blank for every company and the axis dropped that year.	Showed items with no data on a categorical axis and added a note on the page.	A blank first period is a data fact, not an error; explain it.
Chart titles were unclear	Default titles such as "Selected Metric Value by company" did not say what was shown.	Wrote clear titles, with DAX title measures that change with the selected metric or company.	A title should say what is shown, for whom and in which unit.
Theme colours not applied to some visuals	Formatting set by hand on individual visuals overrides a report theme.	Imported the CVI Dark Executive theme and reset those visuals to default before final styling.	Use a theme for consistency; manual overrides defeat it.
Company slicer tiles had no classification colour	Slicer tiles use a single colour by default.	Added conditional-formatting measures (background and text) based on Classification, with a solid colour for the selected tile.	Use colour consistently so one meaning carries across every page.
Operating-income card showed 62.51K instead of +56.59M	The card used the raw operating income field instead of the per-	Built a Sel Op Income Change measure from operating_income_change_usd_mn.	Use measures that follow the

	company change measure.		slicer selection, not raw columns.

18. End-to-End Architecture

SEC EDGAR/XBRL → Python extraction → reusable ETL → annual filtering/deduplication → standardization → company-year financial dataset → derived metrics & QA → investment intelligence → final screening dataset → MySQL → SQL analysis → candidate classification → Power BI.

19. Current Limitations

- The current model is financial-first and is not complete investment due diligence.
- The directional score is transparent but not a valuation model.
- Valuation, TAM, competition, strategic fit, management quality and industry outlook are not yet included.
- The current screening table is a company-level analytical output.
- The Power BI dashboard is delivered as three pages (Executive Overview, Performance Comparison, Company Detail); the method and limitations are documented here and in the README rather than as a fourth dashboard page.
- Final investment conclusions must distinguish screening evidence from broader investment judgment.
- The classification is currently derived in Power BI with DAX because the MySQL screening table has no classification column; the thresholds should be aligned with the SQL classification logic and stored in the database.
- Two items to review in the notebook/SQL: the Crocs earnings-vs-cash-flow flag reads "Aligned positive" although net margin is −2.01%, and Crocs shows +1.98% revenue growth over FY2023–FY2025 but zero positive signals (the growth signal may use the latest year).
- The universe is five companies and three fiscal years, so the results illustrate the method and are not a market-wide screen.
- Year-on-year revenue growth starts in FY2024 because FY2022 is not in the dataset.

20. Power BI Dashboard — Final Build

20.1 Design approach

- Three HTML prototypes were reviewed before building: an analytics-style tabbed layout, a more creative "Deal Room" layout, and a simple single-page layout. The simple version was chosen because it is the easiest to read and maps directly to standard Power BI visuals (cards, bar charts, slicers, tables).
- A dark executive theme (CVI Dark Executive) was applied for a more sophisticated look; green, amber and red mean Strong, Watchlist and Weak on every page, including the company slicer.
- Power BI Desktop is connected to the MySQL database investment_intelligence; the dashboard is clearly labelled as financial screening and decision support.

20.2 Pages delivered

Page	Content	Status
Executive Overview	Title banner; KPI cards (companies screened, Strong, Watchlist, Weak, positive/negative signals); net-score chart coloured by classification; candidate ranking table; "Reading the result" note	Completed
Performance Comparison	Five-button metric slicer; company comparison column chart (Strong → Weak order, classification colours); year-by-year trend line chart that follows the selected metric; full metric	Completed

	comparison table	
Company Detail	Company slicer coloured by classification; five KPI cards; revenue by year; margins by year; interpretation text	Completed

20.3 Data model & DAX objects

The model has two tables joined many-to-one on financial_data[ticker] → investment_screening[company] with single-direction filtering, so the company slicer filters both the company-level cards and the yearly charts.

Object	Type	Purpose
Net Score	Calculated column	Positive minus negative direction count
Classification	Calculated column	Strong / Watchlist / Weak from Net Score (Strong ≥ +3, Watchlist 0 to +2, Weak < 0)
Company Order	Calculated column	Fixes the company order Strong → Weak on charts and slicers
Classification Color, Op Income Color	Measures	Hex colours for bars, table cells and slicer tiles
Slicer Back Default, Slicer Text Color	Measures	Classification colours for the company slicer tiles
Companies Screened, Strong Candidates, Watchlist, Weak Candidates	Measures	KPI cards (distinct company counts)
Total Positive Signals, Total Negative Signals, Positive / Negative Signals	Measures	Signal totals and the "14 / 11" card
Rank	Measure	Companies with a higher Net Score + 1; ties share a rank
Metric Options	Calculated table	Names and order of the five metric slicer buttons
Selected Metric Value	Measure	SWITCH over the selected button for the company comparison chart
Selected Metric Trend, Selected Trend Title	Measures	Year-by-year trend for the selected metric, and its title
Sel Classification, Sel Net Score, Sel Revenue Growth, Sel Op Income Change, Sel Earnings vs CFO	Measures	Company Detail KPI cards that follow the company slicer
Revenue $M; Gross, Operating, Net and CFO Margin %	Measures (financial_data)	Company Detail charts; margins return blank when the source value is NULL

20.4 Future dashboard enhancements

- Add a Method & Limits page inside the dashboard.
- Store the classification rule in MySQL (a view or column) so SQL and Power BI use one definition.
- Publish to Power BI Service with scheduled refresh.
- Add valuation, market-size and strategic-fit measures once that data is sourced.

21. Interview-Ready Project Explanation

I built a source-backed corporate venture screening workflow using SEC EDGAR/XBRL data for five public companies across FY2023–FY2025. I first validated the extraction approach on Etsy and then generalized it into reusable Python ETL with annual filtering, deduplication, standardization and QA. I derived growth, profitability and cash-flow indicators and converted their direction into transparent screening signals. I stored the result in MySQL and used SQL to compare companies, calculate a net-direction score and classify them into Strong Candidate, Watchlist and Weak Candidate groups. Duolingo and Shake Shack emerged as the strongest financial-screening candidates, Boot Barn as a Watchlist name, and Etsy and Crocs as weak candidates under the current rules.

I then built a three-page Power BI dashboard on the MySQL data: an executive overview, a metric-switching performance comparison with a year-by-year trend, and a company detail page. I agreed the layout through three prototypes, chose the simplest, and applied a dark executive theme with one colour meaning (green, amber, red) across every page. I diagnosed and documented the issues along the way, such as table naming, a missing classification column, ranking ties, a broken field parameter, a partial MySQL import and missing slicer relationships, and fixed each one. The project is a screening tool for deciding where to spend due-diligence time, not an investment recommendation.

21.1 Key learnings

- Reconcile every KPI against the SQL results before trusting a visual.
- Keep business rules, such as classification, in the database, not only in the dashboard.
- Show missing data as missing, never as zero, and explain it on the page.
- Test ranking logic with ties.
- Prefer simple, controllable designs (a small table plus one SWITCH measure) over clever ones that are hard to debug.
- Check keys and relationships in the data model before building visuals.
- Match the chart type to the question, and write titles that say what is shown.
- Prototype the layout first, and use one colour meaning across every page.

22. Documentation Maintenance

- This is Version 2.0 (Final) of the project documentation.
- Future work should be added to the relevant existing section instead of creating disconnected notes.
- Update the roadmap whenever a phase changes status.
- Record meaningful technical errors and resolutions in the Error & Resolution Log.
- Keep financial screening conclusions separate from broader investment recommendations.
- Record Power BI and DAX issues and resolutions in the same Error & Resolution Log.
- Any future phase (broader investment factors) should be added to the roadmap as a new phase and integrated into the relevant sections.

23. Project Completion Checkpoint

Project complete: the Python ETL runs end-to-end, financial screening is validated, results are stored in MySQL and analysed in SQL, and a three-page Power BI dashboard (Executive Overview, Performance Comparison, Company Detail) presents them with consistent classification colours. Duolingo and Shake Shack are the strongest financial-screening candidates; Boot Barn is a Watchlist name; Etsy and Crocs are weak under the current rules. Future scope: valuation, market size, competition, strategic fit and management quality.